

2017 H2 Q15

Section: Electricity

Topic: Resistivity — $R = \rho L/A$ and R vs L graph

(a) Calculate R for an aluminium wire

Given: $\rho = 2.8 \times 10^{-8} \Omega \text{ m}$, $L = 0.82 \text{ m}$, $A = 4.0 \times 10^{-6} \text{ m}^2$.

Use $R = \rho L / A$.

$$R = (2.8 \times 10^{-8} \times 0.82) / (4.0 \times 10^{-6}) = 5.74 \times 10^{-3} \Omega.$$

Answer: $R \approx 5.74 \text{ m}\Omega$.

(b) Determine the resistivity of a metal wire from R vs L data

Cross-sectional area $A = 4.52 \times 10^{-6} \text{ m}^2$.

$L \text{ (m)}$	$R \text{ (}\times 10^{-3} \Omega\text{)}$
1.5	5.6
2.0	7.5
2.5	9.4
3.0	11.2
3.5	13.2

(i) Plot R (in Ω) on the y-axis against L (in m) on the x-axis; draw a best-fit straight line.

(ii) From the graph, determine the gradient $m = \Delta R / \Delta L$ (units: $\Omega \text{ m}^{-1}$).

(iii) Resistivity: since $R = \rho L / A$, the gradient $m = \rho / A$, so $\rho = m A$.

Using a least-squares fit to the five data points:

Gradient $m \approx 0.00378 \, \Omega \, \text{m}^{-1}$; intercept $\approx -7.0 \times 10^{-5} \, \Omega$ (close to 0, as expected).

$\rho = m A = 0.00378 \times 4.52 \times 10^{-6} = 1.71 \times 10^{-8} \, \Omega \, \text{m}$.

Answer: $\rho \approx 1.7 \times 10^{-8} \, \Omega \, \text{m}$.

Revision tips

- For a uniform wire, $R \propto L$; the slope gives ρ/A .
- Quote ρ to 2–3 significant figures with correct units ($\Omega \, \text{m}$).
- Check that the y-intercept is close to zero; a non-zero intercept suggests contact/leads resistance.